

PLD Lab 5

question 1.

you will need 8 AND + 3 OR + 3 NOT = 14 gates.

question 2.

you will need 3 AND + 2 OR = 5 gates.

question 3

S_n will have 14 gates.

C_{n+1} will have 5 gates.

1 adder would need $14 + 9 = 19$ gates.

4 adders would need $19 \times 4 = 76$ gates.

question 6.

I think yes that $A_n \oplus B_n$ can be shared. It would make the total gates 2 XOR, 2 AND and 1 OR = 5 gates for each full adder.

A 4 bit 4 full adder would have 20 gates in total.

question 7.

No. As the SOP require a certain amount of gates. In a logic expression stand point, XOR is often use to help reduce the complexity and number of gates.

question 4:

proof: $S_n = A_n \oplus B_n \oplus C_n$

$$S_n = A'_n B_n C_n + A_n B'_n C'_n + A_n B'_n C_n + A_n B_n C'_n$$

↳

1st distribute Law

$$A'_n (B'_n C_n + B_n C'_n) + A_n (B'_n C_n + B_n C'_n)$$

↳ XOR def.

$$A'_n (B_n \oplus C_n) + A_n (B_n \oplus C_n)$$

↳ XOR def

$$\rightarrow A_n \oplus B_n \oplus C_n$$

the exclusive OR def.

is needed for this work.

question 5.

$$C_{n+1} = A_n B_n + A_n C_n + B_n C_n$$

↳

1st. distri. Law

$$A_n B_n + C_n (A_n + B_n)$$

↳ minterm expanding

$$A_n B_n + C_n (A'_n B_n + A_n B'_n + A_n B_n) \quad \text{1st distri. Law}$$

$$A_n B_n + C_n A'_n B_n + C_n A_n B'_n + C_n A_n B_n$$

↳ Absorption Law

$$A_n B_n + C_n A'_n B_n + C_n A_n B'_n$$

↳ 1st distri. Law

$$A_n B_n + C_n (A'_n B_n + A_n B'_n)$$

$$C_{n+1} \rightarrow A_n B_n + C_n (A_n \oplus B_n) \quad \text{XOR def}$$

There are 1 OR, 1 XOR and 2 AND = 4 gates.

question 9.

range of 2's complement values.

$$\text{min.} = -8 (1000)$$

$$\text{max.} = 7 (0111)$$

$$[-8 \text{ to } 7]$$